

Active set method for convex quadratic problem (0.75 pt)

Consider the following 2-dimensional minimization problem

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 \\ \text{subject to:} \quad & x-2y+2 \geq 0, \quad (C_1) \\ & -x-2y+6 \geq 0, \quad (C_2) \\ & -x+2y+2 \geq 0, \quad (C_3) \\ & x \geq 0, \quad (C_4) \\ & y \geq 0 \quad (C_5) \end{aligned}$$

use the active set method for convex QPs starting from the following points:

1. $(x^0, y^0) = (1, 1/2)^T$ for $\mathcal{A}^0 = \emptyset$
2. $(x^0, y^0) = (3, 1/2)^T$ for $\mathcal{A}^0 = \{3\}$
3. $(x^0, y^0) = (0, 1/2)^T$ for $\mathcal{A}^0 = \{4\}$

Sketch the feasible set and for every starting point:

1. Sketch the path of the active set method.
 2. Verify the path of the active set method computationally.
- Note for each step the following states in the table below:

- the primal variables before the step (x_k, y_k)
- the active set before the step A_k
- the primal variables after the step (x_{k+1}, y_{k+1})
- the dual variables after the step λ_{k+1} , e.g. $(0, 1, -1, 0, 0)$
- the active set after the step A_{k+1}
- whether the KKT conditions are satisfied for $(x_{k+1}, y_{k+1}, \lambda_{k+1})$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	$(1, 1/2)$	$\emptyset$				
1						
2						
3						

[Hint: See Nocedal example 16.4 p. 475. and exercise 16.10]

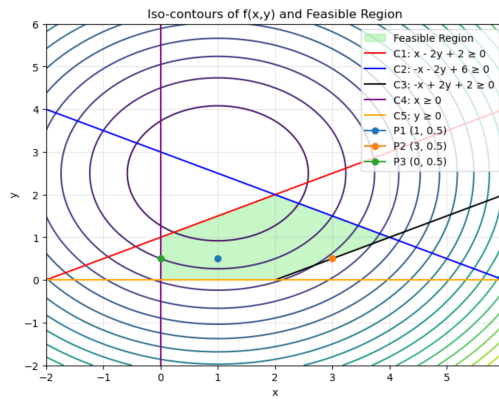
First, I will rewrite the optimization problem with the inequality constraints in standard form $(c_i \leq 0)$.

$$\begin{aligned} \min \quad & (x-1)^2 + (y-2.5)^2 \\ \text{subject to} \quad & -x+2y-2 \leq 0 \quad (c_1), \\ & x+2y-6 \leq 0 \quad (c_2), \\ & x-2y-2 \leq 0 \quad (c_3), \\ & -x \leq 0 \quad (c_4), \\ & -y \leq 0 \quad (c_5) \end{aligned}$$

I will also write each constraint in the form $a_j^T x \leq b_j$:

$$\begin{aligned} a_1 &= \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad b_1 = 2 & a_4 &= \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad b_4 = 0 \\ a_2 &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad b_2 = 6 & a_5 &= \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad b_5 = 0 \\ a_3 &= \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad b_3 = 2 \end{aligned}$$

Here we have the visualization of the optimization problem:



$f(x) = (x-1)^2 + (y-2.5)^2$ can be written in QP form $\frac{1}{2}z^T Q z + q^T z + \text{const}$

using: $Q = 2I_2$ and $q = \begin{bmatrix} -2 \\ -5 \end{bmatrix}$.

Also $\nabla f(x) = [2(x-1), 2(y-2.5)]$.

Let's apply Active-set method for each case:

1. $(x^0, y^0) = (1, 1/2)^T$ for $A^0 = \emptyset$

ITER 0

1) Solve KKT system $\begin{bmatrix} Q & B_0^T \\ B_0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_0 \end{bmatrix}$

As $A^0 = \emptyset$ B_0 and b_0 are 0:

$$Q \cdot x_1 = -q \Rightarrow x_1 = -Q^{-1}q = -\frac{1}{2}q = -\frac{1}{2} \begin{bmatrix} -2 \\ -5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2.5 \end{bmatrix}$$

2) Check for inactive blocking constraints.

C_1 and C_2 are violated because $a_1^T x_{min} > b_1$ and $a_2^T x_{min} > b_2$.

Step lengths: $\alpha_j = \frac{b_j - a_j^T x_k}{a_j^T (x_{k+1} - x_k)}$

$$\alpha_1 = \frac{2 - [-1 \ 2] \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}}{[-1 \ 2] \left(\begin{bmatrix} 1 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \right)} = \frac{2 - 0}{[-1 \ 2] \begin{bmatrix} 0 \\ 2 \end{bmatrix}} = \frac{2}{4} = 0.5$$

$$\alpha_2 = \frac{6 - [-1 \ 2] \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}}{[-1 \ 2] \begin{bmatrix} 0 \\ 2 \end{bmatrix}} = \frac{6 - 2}{4} = 1$$

C_1 is the first blocking constraint.

3) Update A_1 and x_1 :

$$x_1 = x_0 + \alpha_1 (x_1 - x_0) = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} + 0.5 \left(\begin{bmatrix} 1 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A_1 = A_0 \cup \{1\} = \{1\}$$

ITER 1

1) Solve KKT system $\begin{bmatrix} Q & B_1^T \\ B_1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_1 \end{bmatrix}$

$$\begin{cases} Qx_2 + \begin{bmatrix} -1 \\ 2 \end{bmatrix} \lambda = -q \\ [-1 \ 2] x_2 = b_1 \end{cases} \Rightarrow \begin{cases} 2x_2^1 - \lambda = 2 \\ 2x_2^2 + 2\lambda = 5 \\ -x_2^1 + 2x_2^2 = 2 \end{cases} \xrightarrow{\text{Using online solver}} x_2 = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix} \quad \lambda = 0.8$$

2) There aren't blocking constraints.

3) $A_2 = A_1$

KKT are satisfied. k1) All constraints are satisfied

k2) $\lambda_i > 0$

k3) $\lambda_{C1} = 0.8 > 0$

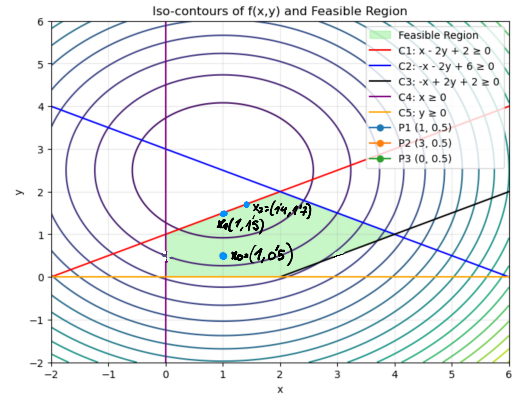
k4) $\nabla f(x^*) + 0.8 [-1 \ 2] = [2(1.4-1), 2(1.7-2.5)] + 0.8 [-1 \ 2] = 0$

ITER 3

$x_3 = x_2$ so the algorithm stops.

$$x^* = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix}$$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(1, 1/2)	$\emptyset$	(1, 1.5)	-	$\{1\}$	NO
1	(1, 1.5)	$\{1\}$	(1.4, 1.7)	0.8	$\{1\}$	NO for x_1
2	(1.4, 1.7)	$\{1\}$	(1.4, 1.7)	0.8	END	YES
3						



2. $(x^0, y^0) = (3, 1/2)^T$ for $A^0 = \{3\}$

ITER 0

1) Solve KKT system $\begin{bmatrix} Q & B_0^T \\ B_0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_0 \end{bmatrix}$

$$\begin{cases} Qx_1 + \lambda \begin{bmatrix} 1 \\ -2 \end{bmatrix} = -\begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ \begin{bmatrix} 1 & -2 \end{bmatrix} x_1 = 2 \end{cases} \Rightarrow \begin{cases} 2x_1 + \lambda = 2 \\ 2x_1 - 2\lambda = 5 \end{cases} \xrightarrow{\text{using online solver}} x_1 = \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} \lambda = -2.4$$

2) There aren't blocking constraints.

3) $A_1 = A_0$

KKT are NOT satisfied. k1) All constraints are satisfied ✓
k2) $\lambda_1 > 0$ XX

ITER 1

1) Solve KKT system $\begin{bmatrix} Q & B_1^T \\ B_1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_1 \end{bmatrix}$

Same as previous: $x_2 = x_1$, $\lambda_2 = \lambda_1$.

Therefore $A_2 = \emptyset$

ITER 2

1) Solve KKT system $\begin{bmatrix} Q & B_2^T \\ B_2 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} -q \\ b_2 \end{bmatrix}$

From previous case we know that $x_3 = \begin{bmatrix} 1.4 \\ 2.5 \end{bmatrix}$ $\lambda_3 = 0.8$

2) The most blocking constraint is C_1 .

$$\alpha_j = \frac{b_j - \alpha_j^T x_k}{\alpha_j^T (x_{k+1} - x_k)} \quad \alpha_1 = \frac{2 - \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix}}{\begin{bmatrix} 1 & 2 \end{bmatrix} \left(\begin{bmatrix} 1.4 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} \right)} = \frac{2 + 2}{\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -0.8 \\ 2.4 \end{bmatrix}} = \frac{4}{6} = \frac{2}{3}$$

3) Update A_3 and x_3 :

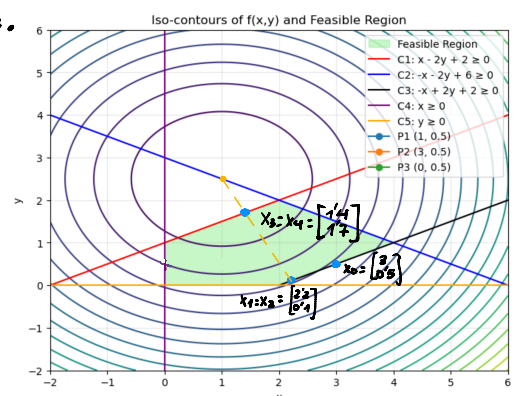
$$x_3 = x_2 + \alpha_1 (x_3 - x_2) = \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} + \frac{2}{3} \left(\begin{bmatrix} 1.4 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 2.2 \\ 0.1 \end{bmatrix} \right) = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix}$$

$$A_3 = A_2 \cup \{1\} = \{1\}$$

From previous case we know that KKT are satisfied for x_3 .

ITER 3 → Stops because $x_4 = x_3$ and $\lambda_4 > 0$.

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(3, 0.5)	$\{3\}$	(2.2, 0.1)	-2.4	$\{3\}$	NO
1	(2.2, 0.1)	$\{3\}$	(2.2, 0.1)	-2.4	$\emptyset$	NO
2	(2.2, 0.1)	$\emptyset$	(1.4, 1.7)	0.8	$\{1\}$	NO for x_2
3	(1.4, 1.7)	$\{1\}$	(1.4, 1.7)	0.8	$\{1\}$	YES



3. $(x^0, y^0) = (0, 1/2)^T$ for $\mathcal{A}^0 = \{4\}$

ITER 0

1) Solve KKT system $\begin{bmatrix} Q & B_0^T \\ B_0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \lambda_1 \end{bmatrix} = \begin{bmatrix} -q \\ b_0 \end{bmatrix}$

$$\begin{cases} Qx_1 + \lambda_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \\ [-1 \ 0] x_1 = 0 \end{cases} \Rightarrow \begin{cases} 2x_1^1 - \lambda_1 = 2 \Rightarrow \lambda_1 = -2 \\ 2x_1^2 = 5 \Rightarrow x_1^2 = 2.5 \\ -x_1^1 = 0 \Rightarrow x_1^1 = 0 \end{cases} \quad x_1 = \begin{bmatrix} 0 \\ 2.5 \end{bmatrix}$$

2) The most blocking constraint is C_1 .

$$\alpha_j = \frac{b_j - a_j^T x_k}{a_j^T (x_{k+1} - x_k)} \quad \alpha_1 = \frac{2 - [-1 \ 2] \begin{bmatrix} 0 \\ 2.5 \end{bmatrix}}{[-1 \ 2] (\begin{bmatrix} 0 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 0 \\ 0.5 \end{bmatrix})} = \frac{2-1}{[-1 \ 2] \begin{bmatrix} 0 \\ 2 \end{bmatrix}} = \frac{1}{4}$$

3) Update A_1 and x_1 .

$$x_1 = x_0 + \alpha_1 (x_1 - x_0) = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} + \frac{1}{4} (\begin{bmatrix} 0 \\ 2.5 \end{bmatrix} - \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A_1 = A_0 \cup \{1\} = \{4, 1\}$$

4) KKT not satisfied because $\lambda_1 < 0$

ITER 1

1) Solve KKT system $\begin{bmatrix} Q & B_1^T \\ B_1 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} -q \\ b_1 \end{bmatrix}$

$$\begin{cases} Qx_2 + \lambda_2^1 \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \lambda_2^4 \begin{bmatrix} -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ [-1 \ 2] x_2 = 2 \\ [-1 \ 0] x_2 = 0 \end{cases} \Rightarrow \begin{cases} 2x_2^1 - \lambda_2^1 - \lambda_2^4 = 2 \Rightarrow \lambda_2^4 = -1.5 - 2 = -3.5 \\ 2x_2^2 + 2\lambda_2^1 = 5 \Rightarrow 2 + 2\lambda_2^1 = 5 \Rightarrow \lambda_2^1 = 1.5 \\ -x_2^1 + 2x_2^2 = 2 \Rightarrow 2x_2^2 = 2 \Rightarrow x_2^2 = 1 \\ -x_2^1 = 0 \Rightarrow x_2^1 = 0 \end{cases}$$

2) As $x_2 = x_1$ and $\lambda_2^4 < 0$.

$$A_2 = A_1 - \{4\} = \{1\}$$

ITER 2

1) Solve KKT system $\begin{bmatrix} Q & B_2^T \\ B_2 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} -q \\ b_2 \end{bmatrix}$

From previous case we know that $x_3 = \begin{bmatrix} 1.4 \\ 1.7 \end{bmatrix}$ $\lambda_3 = 0.8$

2) There are not blocking constraints. x_3 is in C_1 .

3) As in the previous cases, KKT conditions are satisfied

ITER 3 $\rightarrow$ Stops b use $x_4 = x_3$ and $\lambda_4 > 0$.

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	$(0, 0.5)$	$\{4\}$	$(0, 1)$	-2	$\{1, 4\}$	NO
1	$(0, 1)$	$\{1, 4\}$	$(0, 1)$	$\lambda_1^4 = -3.5$ $\lambda_2^1 = 1.5$	$\{1\}$	NO
2	$(0, 1)$	$\{1\}$	$(1.4, 1.7)$	$\lambda_3^1 = 0.8$	$\{1\}$	YES
3	$(1.4, 1.7)$	$\{1\}$	$(1.4, 1.7)$	$\lambda_4^1 = 0.8$	END	YES

